COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 02

BOOLEAN THEOREMS AND GATES



By- Chandan Gupta Sir



Basics of no. system



- Boolean Theorems

Boolean Theorems

$A + B = B + A$

1. Related to 'OR' operation

- $A + 0 = A$
  ('logical OR')
  $\bar{A} + 0 = \bar{A}$
  $AB + 0 = AB$

$1 + f(A, B, C) = 1$

$1 + AB + BC + \bar{A}C = 1$

A	NOT	$\bar{A}$
0		1
1		0

$0 + A = A$

- $1 + A = 1$ ✓✓ $\Rightarrow$ $1 + \text{anything} = 1$

$1 + \bar{A} = 1$

$1 + AB = 1$

$1 + \text{Anything} = 1$

- $1 + 0 = 1$
- $0 + 1 = 1$
- $1 + 1 = 1$
- $0 + 0 = 0$

$f(A, B, C) = A \cdot B \underline{+} BC \underline{+} \bar{A}C$

- $A + A = A$

  $A + A + A = A$

  $A + A + A + A + A = A$

  $\overline{A} + \overline{A} = \overline{A}$

  $\overline{A} + \overline{A} + \overline{A} = \overline{A}$

  $AB + AB = AB$

  $A\overline{B} + A\overline{B} = A\overline{B}$

- $A + \overline{A} = 1$

  $\overline{A} + A = 1$

  $AB + \overline{AB} = 1$

  $x + \overline{x} = 1$

  $AB + \overline{A} \cdot \overline{B} \neq 1$

$\overline{AB} = \overline{A} + \overline{B}$

$\overline{AB} \neq \overline{A} \cdot \overline{B}$

$A \cdot B = AB$

$A\ B$

$A + A + A + 0 + A + A + A = A$

$AB + BC + \overline{A} + \overline{B} =$

$AB + BC + \overline{AB}$

$x + BC + \overline{x} = 1$

$(A + A + 0 + \overline{A} + A + \overline{A}) = 1$

1

2. Related to 'AND' operation

- $A \cdot 0 = 0$ (logical AND)

$0 \cdot A = 0$

$0 \cdot \overline{A} = 0$

$0 \cdot AB = 0$

$(A \cdot B \cdot C) \cdot 0 \cdot D = 0$

0 · anything = 0

$AB = BA$

$A \cdot B = B \cdot A$

- $A \cdot 1 = A$

$1 \cdot A = A$

$\overline{A} \cdot 1 = \overline{A}$

$1 \cdot \overline{A} = \overline{A}$

$A \cdot B \cdot 1 \cdot C = ABC$

- $0 \cdot 0 = 0$
- $0 \cdot 1 = 0$
- $1 \cdot 0 = 0$
- $1 \cdot 1 = 1$

- $A \cdot A = A$

  $A \cdot A \cdot A = A$

  $A \cdot A \cdot A \cdot A = A$

  $\overline{A} \cdot \overline{A} = \overline{A}$

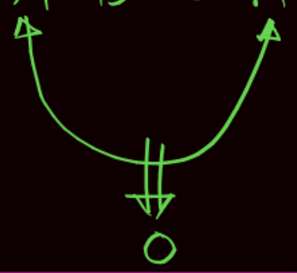
  $\overline{A} \cdot \overline{A} \cdot \overline{A} = \overline{A}$

- $A \cdot \overline{A} = 0$

  $\overline{A} \cdot A = 0$

  $AB \cdot \overline{AB} = 0$

$A \cdot B \cdot C \cdot \overline{A} \cdot D \cdot E = 0$



3. Very imp Boolean theorems

- $A \cdot (B+C) = A \cdot B + A \cdot C$ ⟶ 'AND holds distributive law over OR operation'

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- $A + (B \cdot C) = (A+B) \cdot (A+C)$ ⟶ 'OR holds distributive law over AND operation'

$A + (B \cdot C \cdot D) = (A+B) \cdot (A+C) \cdot (A+D)$

- $A + (\bar{A} B) = (A+\bar{A}) \cdot (A+B) = 1 \cdot (A+B) = (A+B)$

$\bar{A} + (AB) = (\bar{A}+A) \cdot (\bar{A}+B) = 1 \cdot (\bar{A}+B) = (\bar{A}+B)$

- $A + BC = (A+B) \cdot (A+C) = (A + B \cdot C)$

$(A \cdot B) + C$

$C + A \cdot B = (C+A) \cdot (C+B) = (A+C)(B+C)$

$(A+B)(A+\bar{C})(A+\bar{D})$

$= A + (B \cdot \bar{C} \cdot \bar{D})$

Q. $(A+B+\bar{C})(A+\bar{B}+C)(A+\bar{B}+\bar{C})$

$= (A+B+\bar{C})(A+\bar{B})$

$= A + (B+\bar{C})\cdot\bar{B}$

$= A + (0 + \bar{B}\cdot\bar{C})$

$= A + \bar{B}\bar{C}$

$(A+\bar{B}) + C\cdot\bar{C}$

$= (A+\bar{B})$

$[A + (B+\bar{C})\cdot(\bar{B}+C)\cdot(\bar{B}+\bar{C})]$

$[A + (B+\bar{C})[\bar{B} + C\cdot\bar{C}]]$

$= [A + (B+\bar{C})\cdot\bar{B}]$

$= A + \bar{B}\bar{C}$

#Q. $(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})(A+B+\bar{C})$

$[(B+\bar{C})+\bar{A}\cdot A]$

$=(A+\bar{B}+\bar{C})(B+\bar{C})$

$=\bar{C}+(A+\bar{B})\cdot B$

$=\bar{C}+AB$

#Q. $(A+\bar{B}+\bar{C})(A+BC)(\bar{A}+B+C) = A\cdot(B+C) = AB+AC$

$[A+\overline{BC}][A+BC][\bar{A}+B+C]$

$x=BC$

$[A+\bar{x}\cdot x][\bar{A}+B+C]$

$A[\bar{A}+B+C] = A\cdot\bar{A}+A\cdot B+A\cdot C = AB+AC$

Redundancy Theorem :→

- $A \cdot B + \bar{A} C + B \cdot C = A B + \bar{A} C$

redundant term

- $A \bar{B} + \bar{A} \bar{C} + \bar{B} C = (A \bar{C} + \bar{B} C)$

redundant term

$A \bar{x} + x D + A D =$

- $\bar{A} \bar{B} + \bar{B} C + A C = \bar{A} \bar{B} + A C$

redundant term

- $\bar{A} \bar{B} + A C + \bar{B} \bar{C} =$ not applicable

$\bar{A} \bar{B} + A \bar{C} + \bar{B} \bar{C} \Longrightarrow$ N.A.

- $A \bar{B} + A \bar{C} + B C D + A D =$
- $A(\bar{B} + \bar{C}) + B C D + A D$

$A \overline{BC} + B C \bar{D} + A D = A \overline{BC} + B C D$

redundant

- $A\bar{B} + BCD + ACD = (A\bar{B} + BCD)$

redundant

DeMorgan Theorem

- $\overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$

$\overline{A \cdot B \cdot C} = \overline{A} + \overline{B} + \overline{C}$

$\overline{AB + CD + BC} = \overline{A \cdot B} \cdot \overline{CD} \cdot \overline{BC}$

$= (\overline{A}+\overline{B}) \cdot (\overline{C}+\overline{D}) \cdot (\overline{B}+\overline{C})$

$\overline{(A+B) \cdot (C+D)} = \overline{A+B} + \overline{C+D}$

$= \overline{A} \cdot \overline{B} + \overline{C} \cdot \overline{D}$

$\overline{(\overline{B}+\overline{C}) \cdot (C+D)}$

$= \overline{(\overline{B}+\overline{C})} + \overline{C+D}$

$= B \cdot C + \overline{C}\,\overline{D}$

H.W:

Q.1: $A\bar{B} + A\bar{C} + BC$ simplifies to

a. $(A+B)(A+C)$

b. $(A+\bar{B})(A+\bar{C})$

c. $(\bar{A}+B)(A+\bar{C})$

d. $(A+B)(B+C)$

Q.2: $A + \bar{B} + \bar{C} + AD + \bar{A}B$ simplifies to

a. 1

b. $\bar{A} + B + C + D$

c. 0

d. $(A+\bar{B}+\bar{C}+D)$

Q.3: $(A+B)(B+\bar{C})(A+B+C)$ simplifies to

a. $A+B\bar{C}$ b. $B+A\bar{C}$ c. $C+A\bar{B}$ d. $AB+C$.

Q.4: $(A+B\bar{C})(A+C)(A+B+\bar{C})$

equal to

a. A b. $(A+B)(A+\bar{C})$ c. $A+BC$ d. None of the.



2 Minute Summary

→ Basics of Theorems.

ChandanguptaSirpw

Thank you

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Soldiers !

